https://arxiv.org/pdf/2512.03992

Hanwen Wan, Zexin Lin, Yixuan Deng, and Xiaoqiang Ji, computer-vision and artificial-intelligence researchers affiliated with The Chinese University of Hong Kong, Shenzhen, and its Institute of Artificial Intelligence and Robotics for Society, wrote the research preprint “Value-Guided Iterative Refinement and the DIQ-H Benchmark for Evaluating VLM Robustness,” first published on arXiv in 2025. The article investigates how vision-language models respond to degraded visual inputs, adversarial conditions, value misalignment, and errors that persist across continuous sequences. The authors introduce the DIQ-H benchmark to measure these weaknesses and propose a Value-Guided Iterative Refinement framework that automatically improves the accuracy and ethical alignment of generated annotations. Their experiments report an accuracy increase from 72.2% to 83.3%, demonstrating the framework’s potential to support safer and more reliable embodied-AI systems.

https://arxiv.org/pdf/2603.09512

Chun-Peng Chang, Chen-Yu Wang, Holger Caesar, and Alain Pagani, computer-vision and artificial-intelligence researchers affiliated with DFKI Augmented Vision and TU Delft, wrote the research preprint “Probing the Reliability of Driving VLMs: From Inconsistent Responses to Grounded Temporal Reasoning,” published on arXiv in 2026. The article examines whether vision-language models used as driving assistants can consistently interpret traffic scenes and predict how those scenes will develop over time. The authors reveal that these models often produce unstable, contradictory, or poorly timed predictions, even when they demonstrate strong visual understanding. To address these weaknesses, they introduce the human-annotated FutureVQA benchmark and propose a self-supervised, chain-of-thought tuning method intended to improve temporal reasoning and reliability in autonomous-driving applications.

https://arxiv.org/pdf/2405.05885

Ziang Guo, Zakhar Yagudin, Artem Lykov, Mikhail Konenkov, and Dzmitry Tsetserukou, researchers in robotics, autonomous driving, and artificial intelligence, wrote the IEEE conference research paper “VLM-Auto: VLM-based Autonomous Driving Assistant with Human-like Behavior and Understanding for Complex Road Scenes,” published on arXiv in 2024. The article explores how vision-language models can help autonomous vehicles understand complicated road environments and adjust their driving behavior. The authors address the high computational cost and limited flexibility of existing autonomous-driving systems by presenting VLM-Auto, a system integrated with CARLA and ROS2, along with a fine-tuning dataset. Their experiments demonstrate

strong prediction performance in simulated and challenging real-world conditions, suggesting that VLMs can improve adaptable scene understanding for autonomous driving.

https://arxiv.org/pdf/2501.04003
Shaoyuan Xie, Lingdong Kong, Yuhao Dong, Chonghao Sima, Wenwei Zhang, Qi Alfred Chen, Ziwei Liu, and Liang Pan, researchers affiliated with the University of California, Irvine, Shanghai AI Laboratory, the National University of Singapore, Nanyang Technological University, and the University of Hong Kong, wrote the computer-vision and robotics preprint “Are VLMs Ready for Autonomous Driving? An Empirical Study from the Reliability, Data, and Metric Perspectives,” published on arXiv in 2025. The article evaluates whether vision-language models can provide reliable, visually grounded explanations and decisions for autonomous-driving tasks. The authors introduce DriveBench, a benchmark containing clean, corrupted, and text-only inputs, and use it to show that VLMs often rely on prior general knowledge or textual cues instead of accurately interpreting visual information. They conclude that current evaluation methods can conceal serious reliability problems and propose stronger metrics and corruption-aware techniques to support safer, more interpretable autonomous-driving systems.

https://arxiv.org/pdf/2607.09138
Sandro Papais, Letian Wang, Mudit Jain, Behnaz Rezaei, and Steven L. Waslander, autonomous-driving researchers affiliated with the University of Toronto and Qualcomm Technologies, wrote the robotics research preprint “BeyondSight: Object Permanence for End-to-End Autonomous Driving,” published on arXiv in 2026 and accepted to ECCV 2026. The article examines how autonomous vehicles can continue tracking and reasoning about other road users after they become temporarily hidden by vehicles or infrastructure. The authors introduce BeyondSight, a framework that maintains persistent hypotheses about occluded actors and integrates them into perception, prediction, and planning, along with the nuScenes-Permanence evaluation dataset.

https://arxiv.org/pdf/2502.00843
Yuxin Lin, Mengshi Qi, Liang Liu, and Huadong Ma—computer-vision and artificial-intelligence researchers affiliated with the State Key Laboratory of Networking and Switching Technology at Beijing University of Posts and Telecommunications—wrote the research preprint “VLM-Assisted Continual Learning for Visual Question Answering in Self-Driving,” first published on arXiv on February 2,

2025 and revised on December 7, 2025. The article investigates how vision-language models can learn successive autonomous-driving tasks without catastrophically forgetting previously acquired knowledge. The authors propose a continual-learning framework that combines selective memory replay, knowledge distillation, and task-specific projection-layer regularization to preserve performance across perception, prediction, and planning tasks. Evaluations on the DriveLM dataset demonstrate improvements ranging from 20.11% to 35.16%, indicating that the framework can make visual question-answering systems more adaptable and reliable for autonomous driving.

https://arxiv.org/pdf/2502.13923

Introduced by Alibaba's Qwen Team in 2025, Qwen2.5-VL is an open-source vision-language model family available in 3B, 7B, and 72B parameter versions. It uses a native dynamic-resolution vision encoder, window attention, dynamic frame-rate sampling, and absolute-time multimodal positional embeddings. These features support object grounding, detailed image understanding, long-video comprehension, and temporal event localization. On video benchmarks, Qwen2.5-VL-7B achieved 65.1% on Video-MME without subtitles, 69.6% on MVBench, 65.0% on EgoSchema, and 43.6 mIoU on Charades-STA. The 72B version achieved 73.3%, 70.4%, 76.2%, and 50.9 mIoU, respectively. On general multimodal benchmarks, Qwen2.5-VL-72B achieved 70.2 on MMMU, 88.4% on MMBench-EN-V1.1, and 70.8% on MMStar, demonstrating strong performance across image and video understanding tasks.

https://arxiv.org/pdf/2408.03326

Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li introduced LLaVA-OneVision in 2024 as an open-source multimodal model family available in 0.5B, 7B, and 72B parameter versions. It is designed to support single-image, multi-image, and video understanding within a unified model. Its training approach emphasizes transfer between visual modalities, enabling capabilities such as image-sequence comparison, video-change analysis, and multi-camera video interpretation. In the authors' zero-shot evaluation, LLaVA-OneVision-7B achieved 56.6% on ActivityNetQA and 58.2% on Video-MME, while the 72B version achieved 62.3% and 66.2%, respectively. On image-based benchmarks, the 7B model achieved 81.4% on AI2D and 87.5% on DocVQA, compared with 85.6% and 91.3% for the 72B model. These results demonstrate strong performance across both image and video understanding tasks.